

Ex. No: 4Finite State Automata Based Morphological Analyzer

Implement Finite State Automata Based Morphological Analyzer for the given Datasets

Datasets:

- 1. Brown Corpus
- 2. Reuters Corpus
- 3. Newsgroups Corpus
- 4. Universal Dependencies English Tree Bank

Aim

To implement a Finite State Automata (FSA) based Morphological Analyzer to model morphological processes as state transitions, enabling systematic analysis of word forms, their morphemes, and grammatical categories.

Procedure/Pseudocode

- 1. Define states, alphabet (characters), transition function, start state, and accept states for the FSA.
- 2. Model common English morphological patterns as FSA paths: plural formation (+s/+es), past tense (+ed), progressive (+ing), comparative (+er), superlative (+est).
- 3. Implement the FSA as a state transition table (dictionary of dictionaries in Python).
- 4. For each word in the corpus, simulate the FSA by traversing states character by character.
- 5. On reaching an accept state, extract the recognized morphological analysis: stem + suffix + feature.
- 6. Handle special cases: consonant doubling (run→running), e-dropping (make→making), y-to-i changes (happy→happiness).
- 7. Evaluate against UD Tree Bank gold annotations.

Libraries Used

- 1. nltk – for corpus loading
- 2. automata-lib – for formal FSA implementation
- 3. graphviz – for visualizing FSA state diagrams
- 4. pandas – for result tabulation

Test Cases

- 1. FSA for plural: State 0 (START) → [consonant transitions] → State N → '+s' → ACCEPT (plural noun)
- 2. Input: 'boxes' → FSA traversal: b-o-x → e-s → Accept state: root='box', suffix='+es', feature=NOUN+PL
- 3. Input: 'running' → FSA: r-u-n → doubling rule → n-i-n-g → Accept: root='run', feature=VERB+PROG
- 4. Rejection test: 'xkzw' → FSA reaches no accept state → Unknown morphology

Results

The FSA-based morphological analyzer successfully modeled regular English morphological patterns with deterministic state transitions. It outperformed the rule-based approach in structured pattern recognition and was able to visualize morphological paths. Accuracy on the UD Tree Bank reached approximately 82% for regular inflectional morphology.